Bringing Mock-Call Training In-House

Voice-based advisor training — what was built, tested, and cost-measured

August 2026 · Working prototype, validated against live vendor dashboards

A working in-house voice mock-call trainer has been built and tested — a believable AI customer that advisors practise selling to, driven entirely through a web browser. It delivers the capability currently rented from a vendor at ₹6 per minute for a measured ≈₹1.80 per minute: a ~70% reduction in the marginal cost of every call, verified against live vendor dashboards rather than estimated. The same pipeline scales directly into a full production system.

The current position

The present vendor charges a flat ₹6/minute of call. That rate is fixed and scales linearly with adoption — the more advisors use the tool and the longer they train, the more we pay, with no compounding return. We rent the capability rather than own the underlying pipeline, so the cost is a permanent operating line that only grows.

The economics

Running the pipeline ourselves means paying wholesale API rates directly — speech-to-text, the language model, and text-to-speech — with no vendor markup. Measured cost per minute of call:

Component (per minute of call)	In-house rate
Speech-to-text — Sarvam Saaras v3	₹0.50
Language model — Claude Haiku 4.5 (persona)	₹0.37
Text-to-speech — Sarvam Bulbul v3 (the dominant cost)	₹0.93
In-house total (measured average)	**≈₹1.80**

Costs are best seen per trainee batch. One batch is 20 advisors, each completing 40 practice calls of 20 minutes — 16,000 minutes of training in all:

	Vendor @ ₹6/min	In-house @ ≈₹1.80/min	Saving
Per 20-minute call	₹120	₹36	**₹84**
Per advisor (40 calls)	₹4,800	₹1,440	**₹3,360**
Per trainee batch (20 advisors)	₹96,000	₹28,800	**₹67,200**

These are ballpark figures. The true cost of a minute depends on how much is actually spoken in it — chiefly the characters the AI customer voices (text-to-speech is the dominant driver). The system meters every call and computes its exact cost at run time; ≈₹1.80/min is the measured average and will vary (roughly ±30%) with how talkative a given call is. The vendor's ₹6/min, by contrast, is flat regardless. The batch model above is illustrative — adjust batch size or calls per advisor to actuals, and every additional batch repeats the saving.

How it works — the mechanics to replicate

The prototype is a clean, config-driven pipeline that is also the spine of a production system:

> Browser (WebRTC, push-to-talk, microphone selection) → Pipecat orchestrator → Sarvam Saaras v3 speech-to-text → Claude Haiku 4.5 persona model → Sarvam Bulbul v3 text-to-speech → back to the browser as audio + a live transcript.

Four building blocks — all already working — make it replicable. (1) Personas as data: each customer is a JSON file rendered into the model's instructions, so new profiles are authored, not coded. (2) A usage accountant: every call is metered and priced from an auditable rate file, validated to within a few percent of the vendor dashboards. (3) Session persistence: the full transcript plus a training checklist is saved per call for review. (4) Config-driven throughout: call-length cap, voice, and rates are settings, not hardcoded. On call end, a live cost breakdown is shown.

An added advantage: a head start on live-call QA

Every practice call already produces a structured, machine-readable transcript through the same speech pipeline that real customer calls will use. When the planned live-call QA automation goes live, the transcript format, the checklist and scoring hooks, and the pipeline itself are already shared — so integrating the two is far easier than bolting QA onto a closed vendor. The training tool and the QA roadmap reinforce one another rather than being separate builds.

> Recommendation. Approve a short productionization phase to move this from prototype to a supported internal tool. It converts a growing, flat vendor charge into an owned pipeline at roughly 70% lower marginal cost per call — on the order of ₹67,200 saved per trainee batch (₹96,000 down to ₹28,800), repeating with every batch, plus a running head start on live-call QA integration. Risk is low: the pipeline works today, the per-minute cost is measured against real vendor billing, and the remaining work is incremental.

Rates (measured Aug 2026, validated against Sarvam and Anthropic dashboards): Saaras v3 ₹30/hr; Bulbul v3 ₹30/10k characters; Claude Haiku 4.5 $1 / $5 per million tokens (input / output). Volume model: one trainee batch = 20 advisors × 40 calls × 20 minutes = 16,000 minutes. Per-minute cost varies with talk density.